



Data to AI to Large Language Models

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What's the difference between Data, AI, and ML?



What is inside ChatGPT's Brain?



Data → Train → Model → Test →
Predict → Improve



The image features a dark blue background. On the left, three horizontal lines of binary code (0s and 1s) are displayed in light blue, magenta, and yellow. From these lines, a dense web of thin, curved lines in corresponding colors (cyan, magenta, and yellow) flows towards the right. On the right side, there is a large, rectangular area filled with a grid of small, multi-colored dots (blue, green, yellow, magenta) and a background of faint, multi-colored binary code. The word "Data?" is written in a large, white, sans-serif font, centered over the right side of the image.

Data?



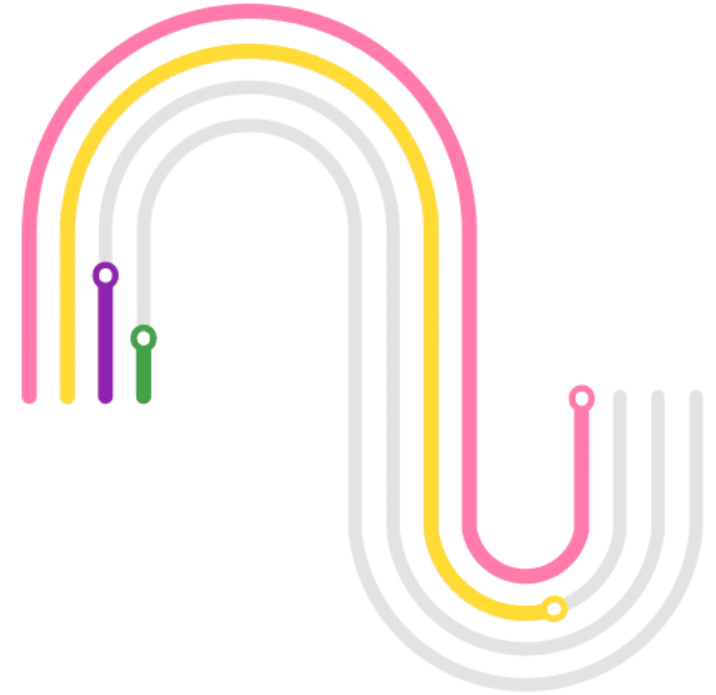
What is Data



Data

- Data **are** raw facts and figures that on their own have no meaning

-
- These can be any alphanumeric characters i.e. text, numbers, symbols





Structured data



Text



Voice



3D signals



Images



Training

Large Language Model



Adaptation



Information
extraction



Instruction
following



Object
recognition



Image
captioning



Q&A



Sentiment
analysis

Data

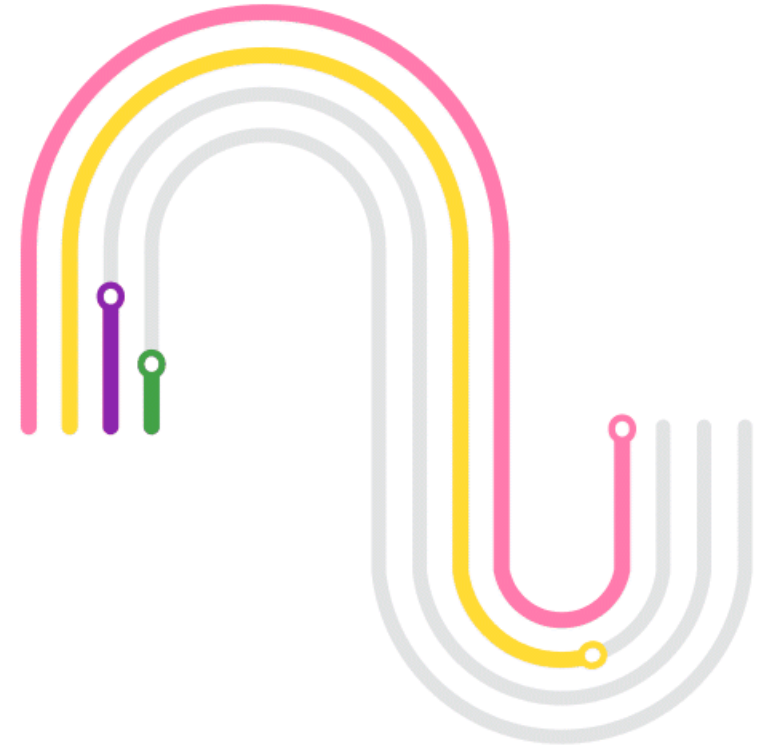
Yes, Yes, No, Yes, No, Yes,
No, Yes

42, 63, 96, 74, 56, 86

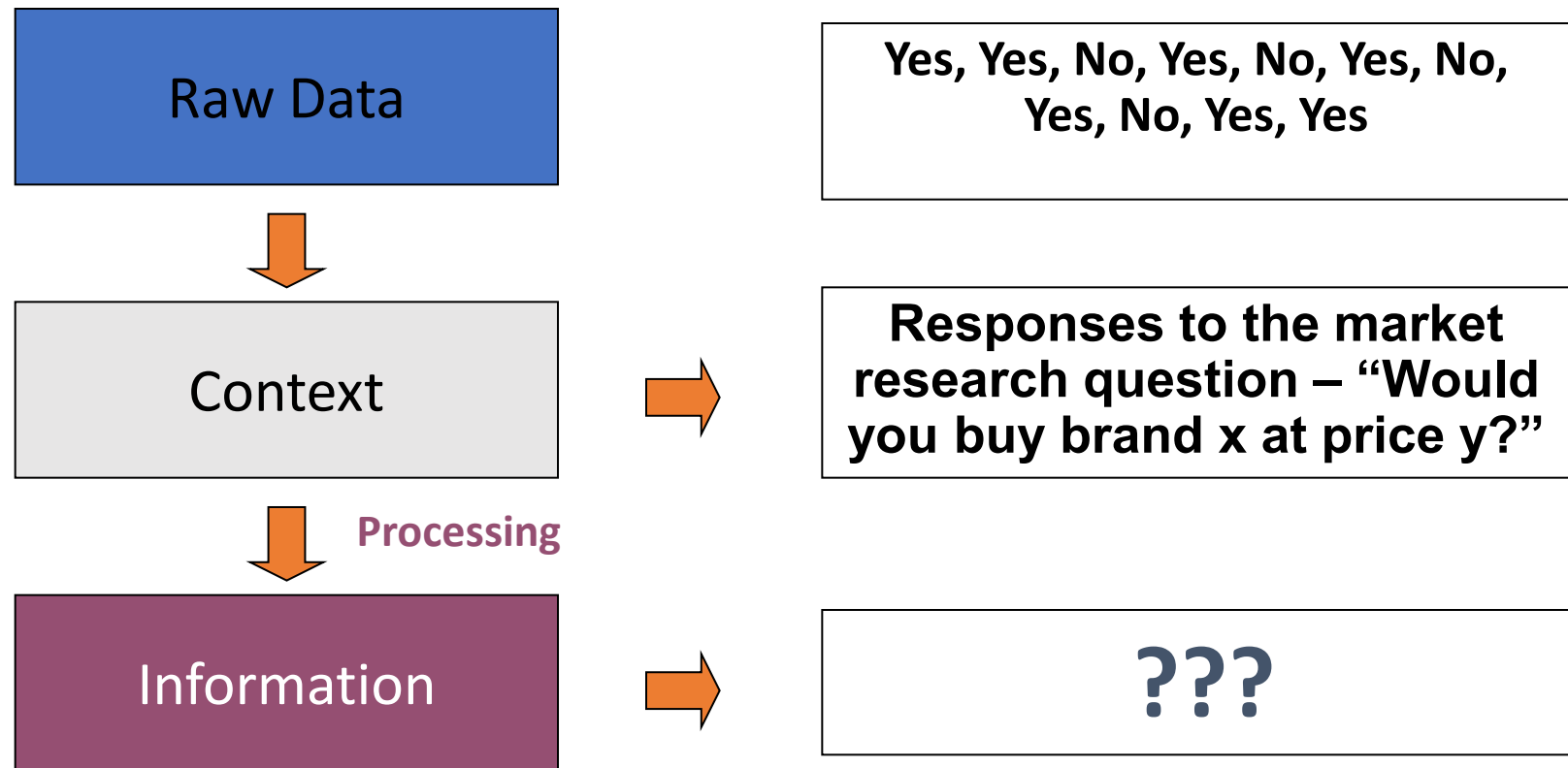
111192, 111234

Number of households owning...

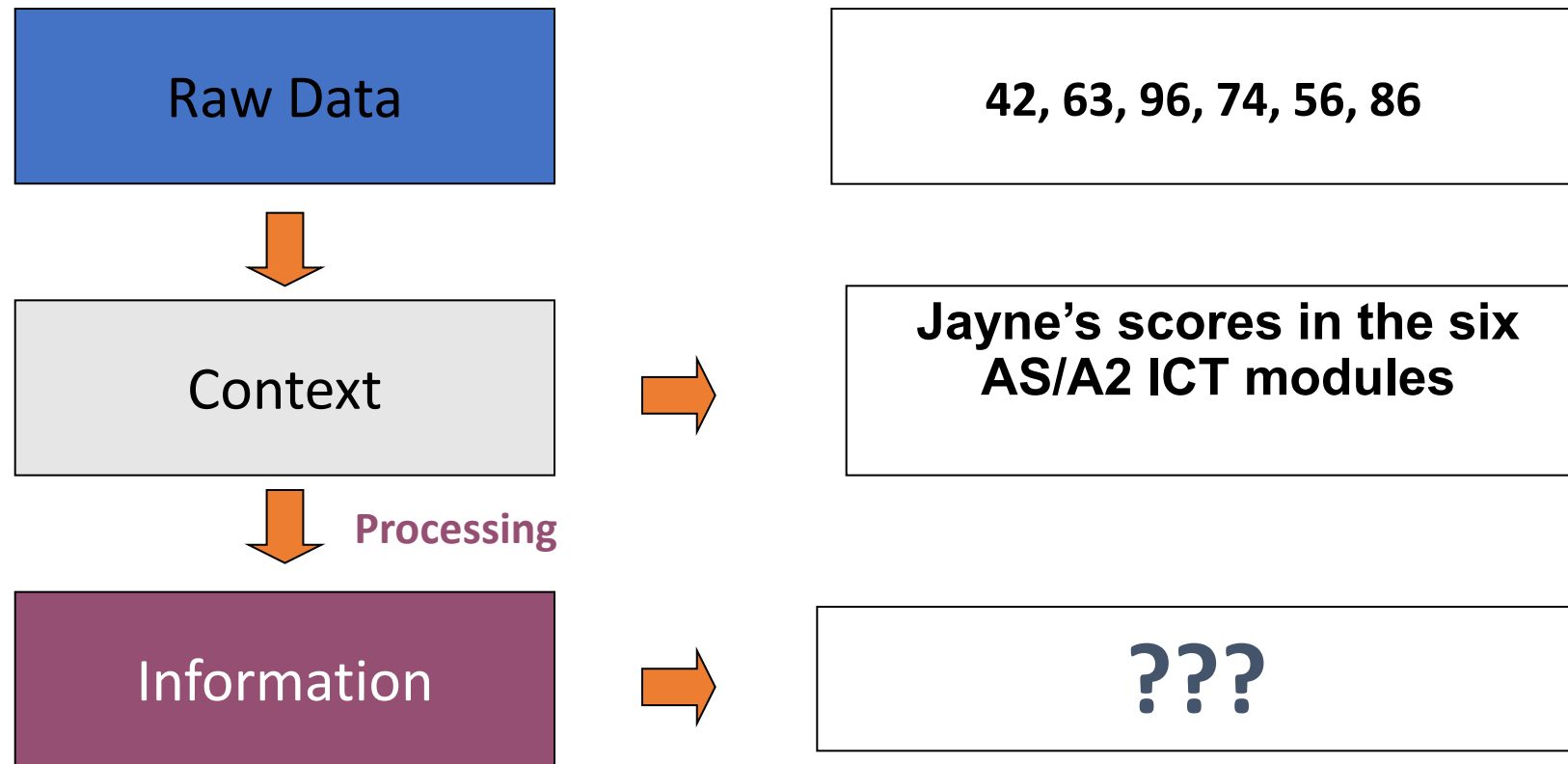
Context



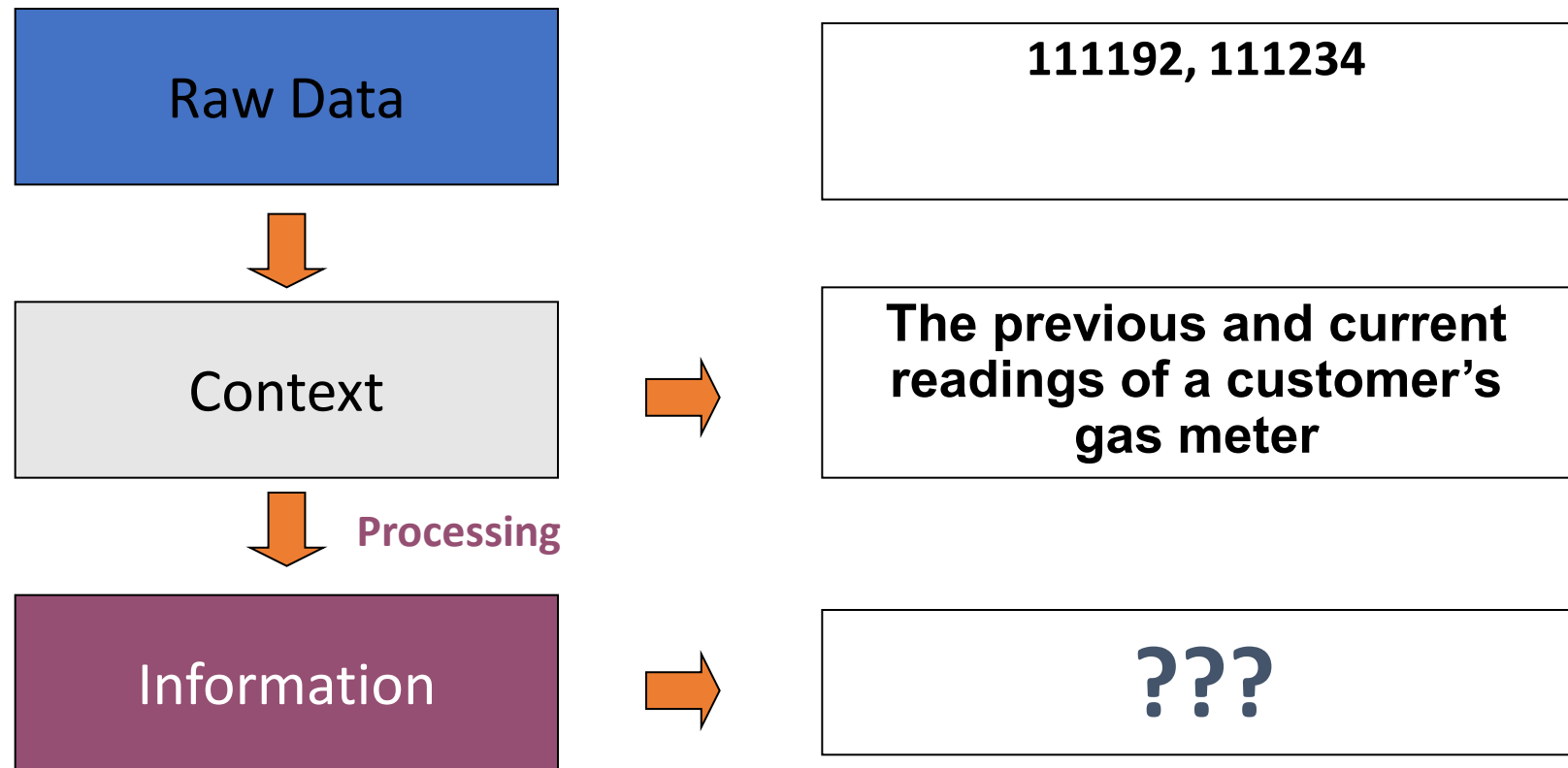
Example 1



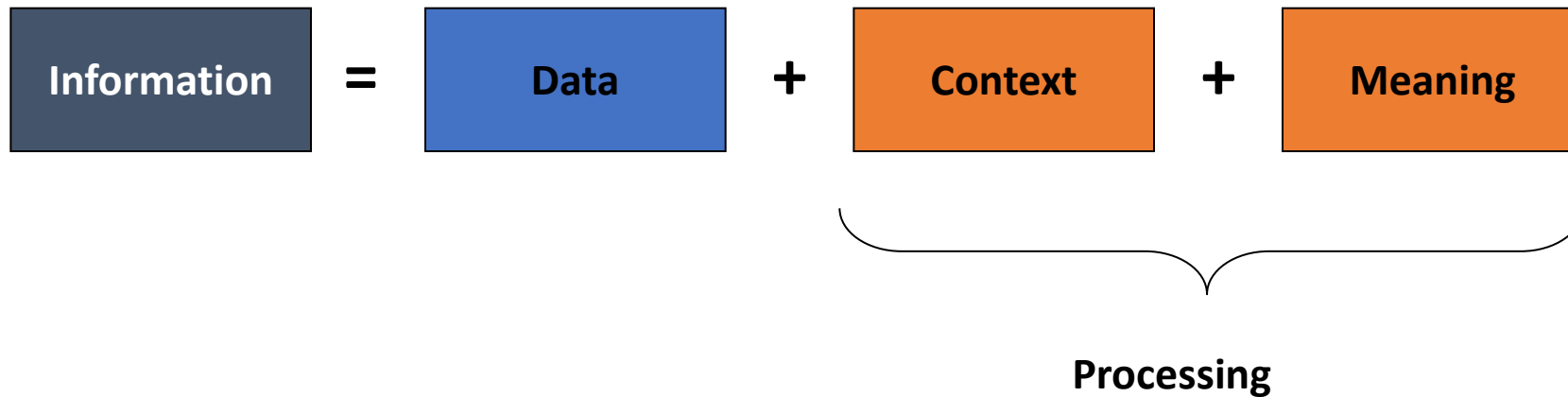
Example 2



Example 3



Summary



Data – raw facts and figures

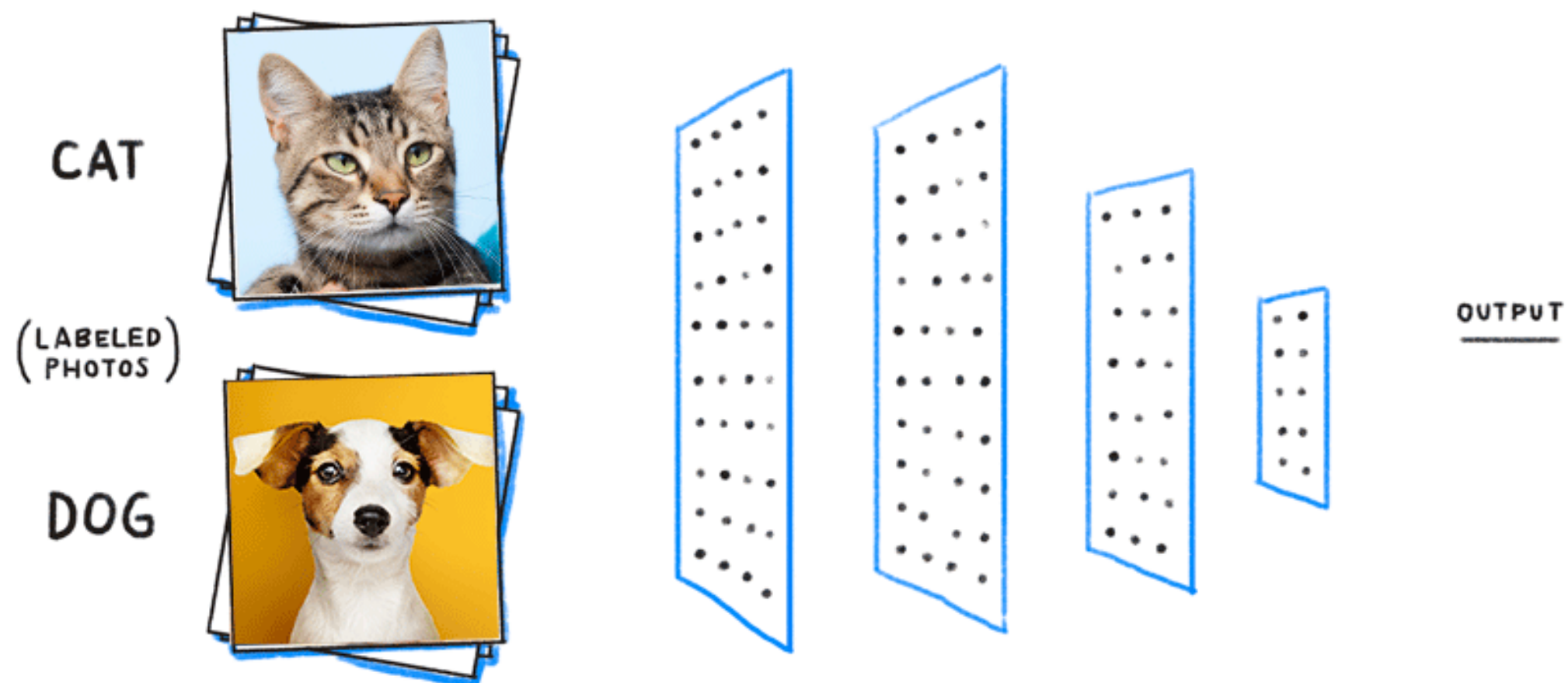
Information – data that has been processed (in a context) to give it meaning



How Machine Learning works



HOW DOES MACHINE LEARNING WORK?



UNSUPERVISED MACHINE LEARNING

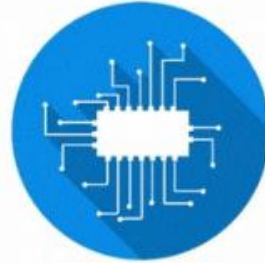
RANDOM DATA



INPUT



PREDICTION



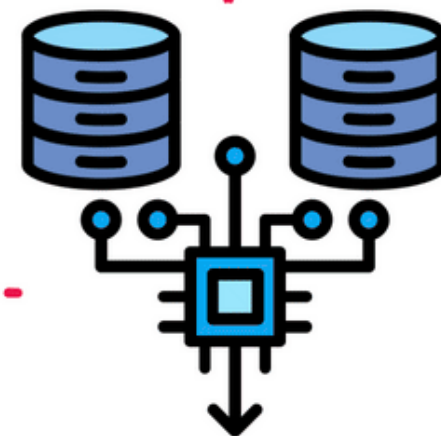
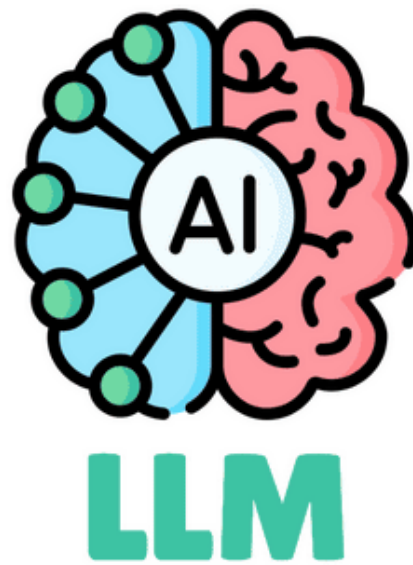
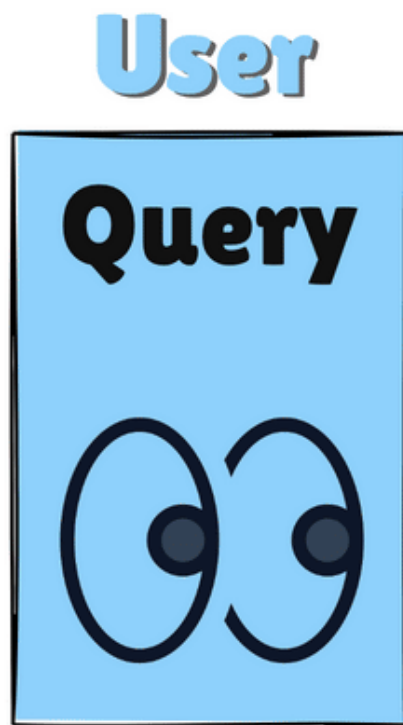
GROUP 1



GROUP 2

How LLMs work?

Do the LLMs Understand?



Training Data



What exactly is a language model?

A language model estimates the probability of a token or sequence of tokens occurring within a longer sequence of tokens. A token could be a word, a subword (a subset of a word), or even a single character.

Large Language Models – Example

- Generate starting from the prompt (initial context) “Hello”
- Each time a new token is appended to the context and the next one is generated

Hello

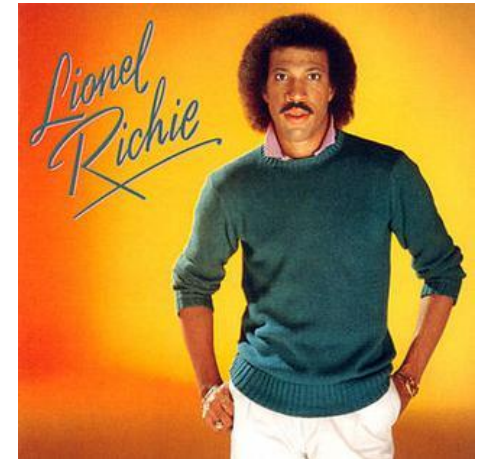
from the other side!

World!

, is it me you're looking for?



[https://en.wikipedia.org/wiki/File:Adele_-_25_\(Official_Album_Cover\).png](https://en.wikipedia.org/wiki/File:Adele_-_25_(Official_Album_Cover).png)



[https://en.wikipedia.org/wiki/File:Lionel_Richie_\(self-titled_album_-_cover_art\).jpg](https://en.wikipedia.org/wiki/File:Lionel_Richie_(self-titled_album_-_cover_art).jpg)

How Do Large Language Models (LLMs) Work?

- At their most basic LLMs are statistical pattern-recognition and prediction systems
- LLMs output the next likely word (“token”) in a sentence (“sequence”)
 - token: unit of text e.g. word, character. 1 word ~ 0.75 token
 - sequence: context - section (“window”) of text e.g. sentence, paragraph, book
 - input into chatGPT is 4096 tokens; Claude 2 is 100K tokens
- The likelihood of the next word appearing is determined by
 - the context in which the words are seen in a larger body of text (“corpus”) and
 - the input to the chat

LLMs “Understand” “Meaning”

- Learning from a large corpus allows LLMs to understand the meaning of words.

For example

- the training data may consist of many sentences beginning with “my favourite colour is...”
 - the next word will be a colour, allowing LLMs to cluster the words “red, blue, green...” into a set that represents the concept of “colour”
-
- It’s important to note that LLMs don’t really understand anything. They create statistical patterns that groups similar tokens based on a complex measure of how similar or dissimilar they are.

Data used to Train LLMs

- LLMs are trained in an unsupervised manner on vast quantities of open source and licensed data e.g.
 - The Pile (825GB, incl. web, papers, patents, books, ArXiv, Stack Exchange, maths problems, computer code)
 - Common Crawl (~20B URLs)
 - GPT3: 175B parameters; GPT4: undisclosed: est. 500B – 1000B - bigger is better (for now, at least)
- Responses are refined using question-response pairs (“InstructGPT”) from the web, humans or bootstrapped (i.e. the LLM outputs its own pairs)
- reinforcement learning with human feedback (RLHF) is used to reward LLMs to give appropriate responses (“guardrails”)
- “Constitutional AI” – trained to filter responses based on e.g. Universal Declaration of Human Rights (Claude 2)

Next Word Prediction

- is influenced by the frequency the word is seen in various contexts
- but there is a degree of randomness so that the word with the highest probability isn't always seen

My favourite colour is

green	9.7%
red	15%
pink	11.6%
puce	2.3%

Beware!

- LLMs may seem to “lie” and “hallucinate” i.e. give what are factually-incorrect responses to questions*
 - as you now know, they’re not trained to do give you an objectively correct answer!
- this is some function of training data (e.g. bias), learning, search and probability
- don’t believe the outputs – they always need checking, at least for now

* LLMs aren’t people. They have no intentionality. Don’t anthropomorphise them 😊

Zero-Shot Prompting

*No examples or context provided, the LLM responds using **only its pre-trained knowledge***



Few-Shot Prompting

*LLM is guided by **examples** in the prompt to understand task format and improve accuracy.*



Chain-of-Thought Prompting

The LLM is **not reasoning**, generating passages helps improve performance by exploiting intermediate passage's output



Chain-of-Thought (CoT) prompting is a method where the prompt instructs the LLM to generate intermediate reasoning steps before arriving at a final answer.

Can be combined with other prompting strategies!

Prompt

*Sarah left her umbrella at home.
She looked outside and saw dark clouds in the sky.
Should she take a raincoat?
Think step-by-step*



Output

Sarah left her umbrella at home, so she doesn't have it with her.
Dark clouds in the sky often indicate that it might rain soon.
Since she doesn't have an umbrella and it might rain, she may get wet.
Taking a raincoat would help her stay dry.
Answer: Yes, she should take a raincoat.

When to use which strategy?

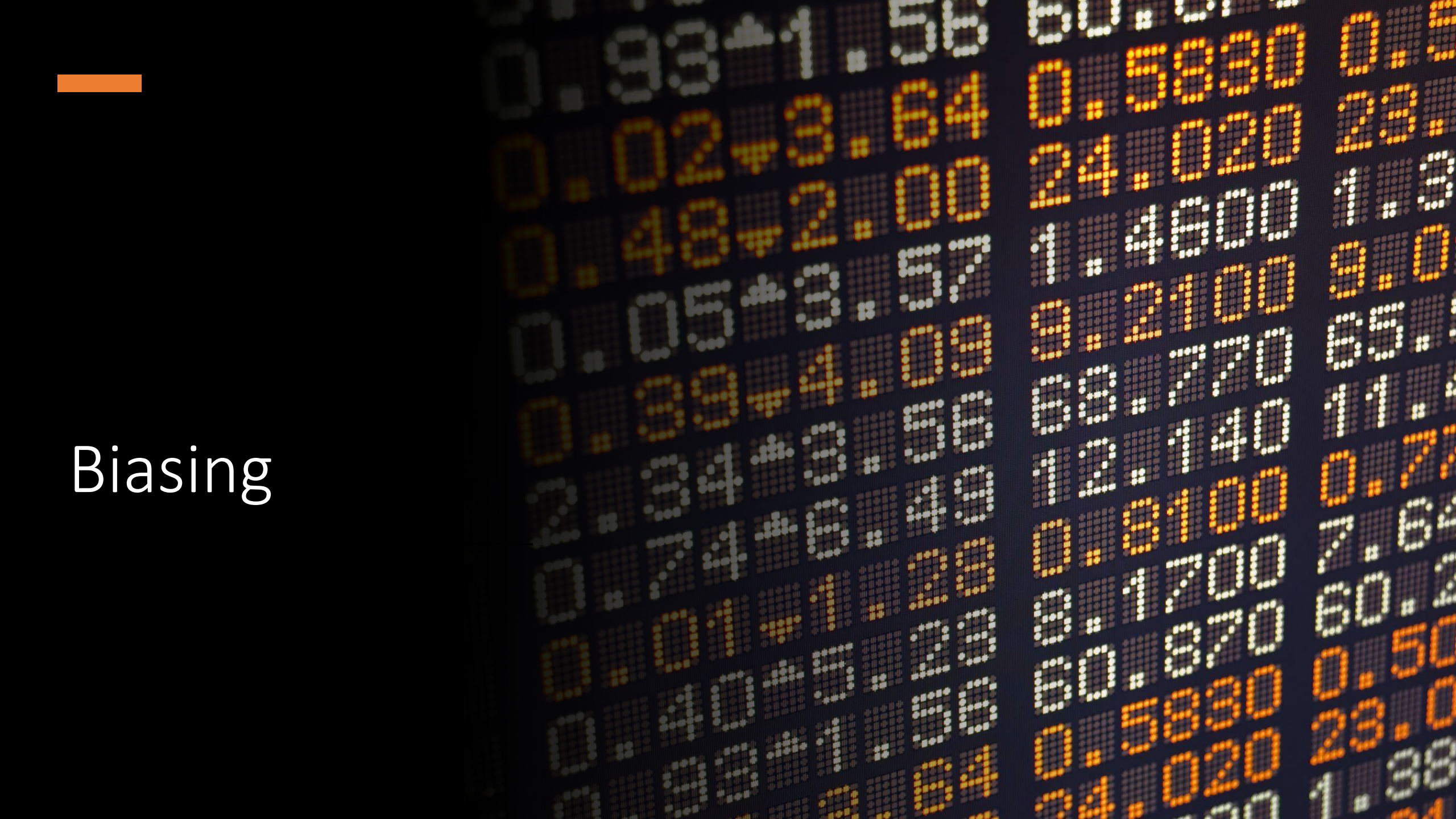
Each strategy improves model performance in different scenarios:

- Use Zero-Shot when **examples aren't feasible** or the task is **simple** or well-known to the model
- Use Few-Shot for clarity, for **domain-specific tasks**, or to teach a specific tone/style/logic
- Use Chain-of-Thought for **complex reasoning tasks** or **multi-step question answering**
- Use RAG for knowledge-intensive tasks and for tasks that require **external or up-to-date knowledge** that the LLM may not have seen during training
- Use GraphRAG, when knowledge is naturally structured (e.g., citations, ontologies, taxonomies), **relationships** between pieces of data matter for accurate answers, or when you need **advanced reasoning** over connected facts

Challenges with AI ML and LLMs



Biasing

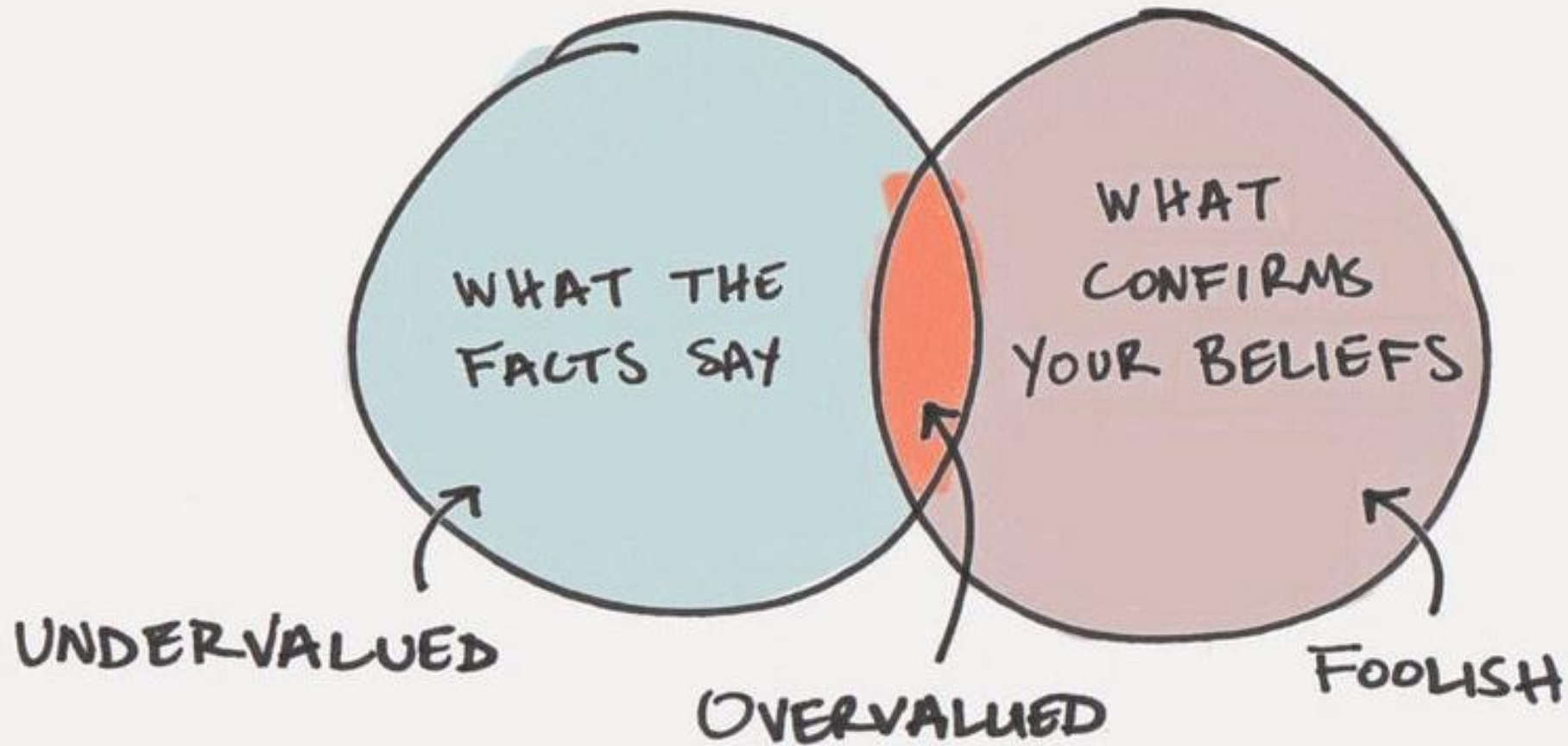


**I THINK JANICE IS NOT A
GOOD TEAM PLAYER.**

**WHAT MAKES YOU
SAY THAT?**

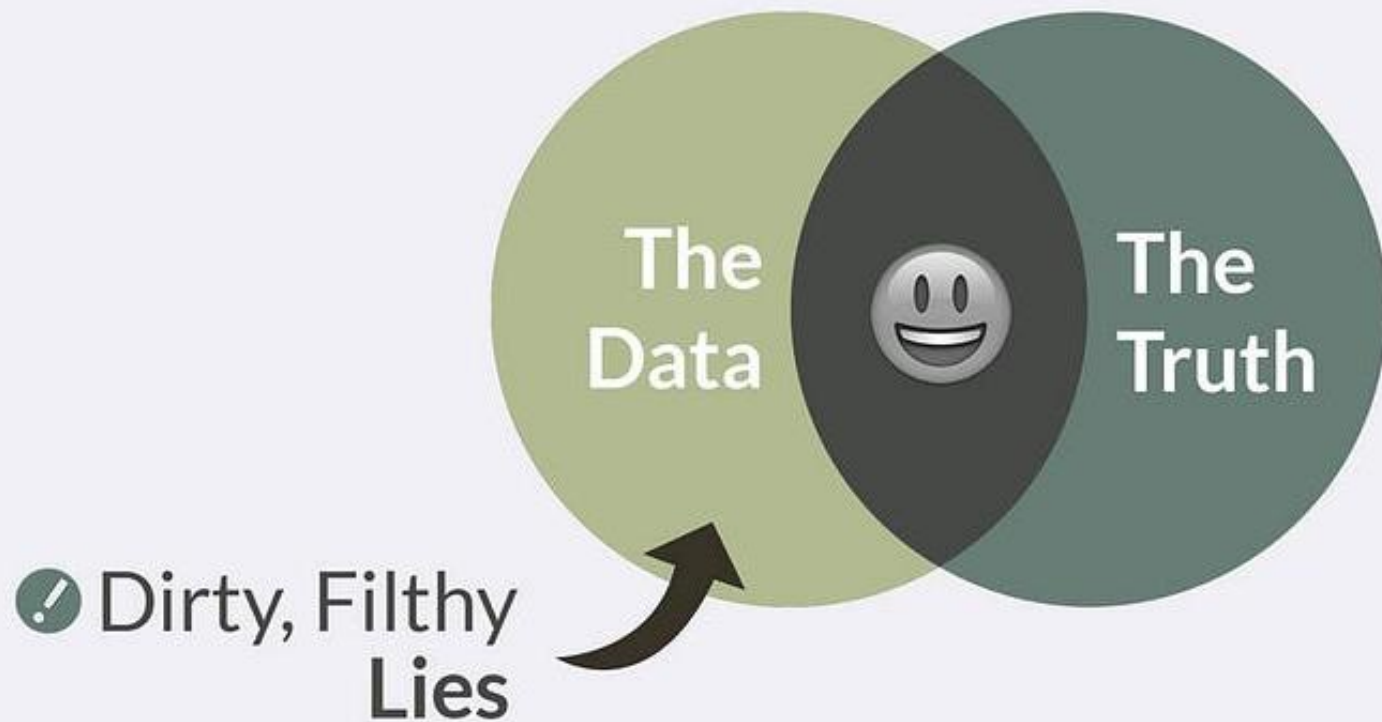
**WELL, I SEE IT ALL THE TIME THAT
SHE DOES NOT MINGLE WELL WITH
HER TEAM MATES.**

THE CONFIRMATION BIAS



The Data ~~Never~~ Lies

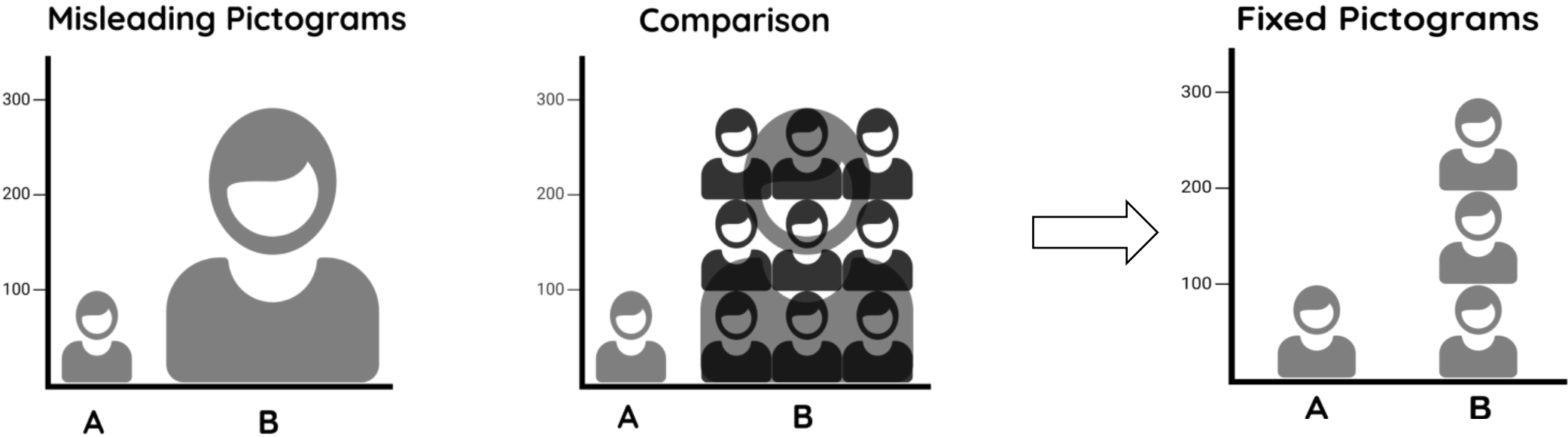
How to Get the Truth from Marketing Data



Data Bias Meaning



How Data Can Mislead

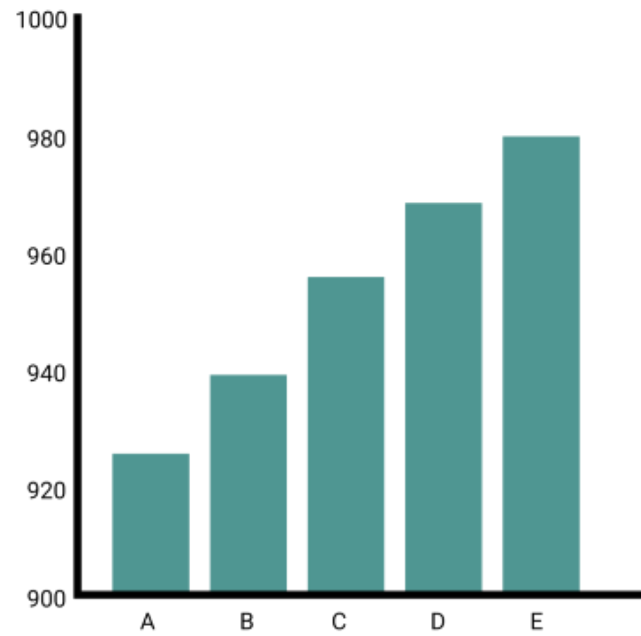


Identifying the root cause a problem?

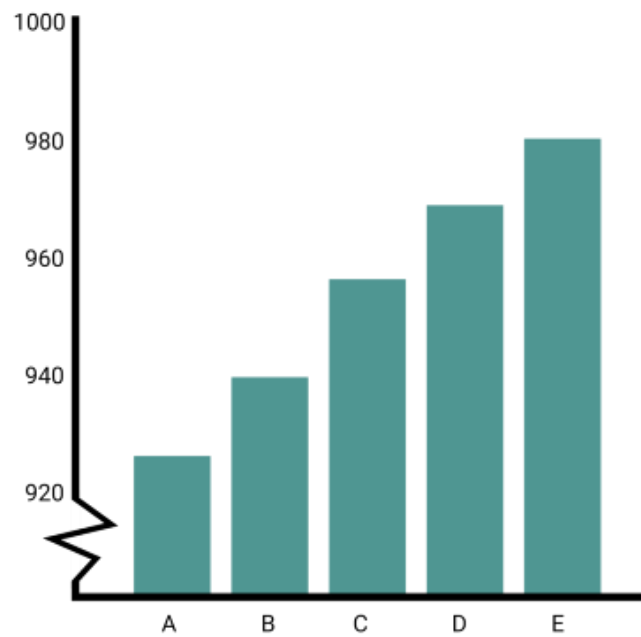


Misleading Graphs

Misleading Graph

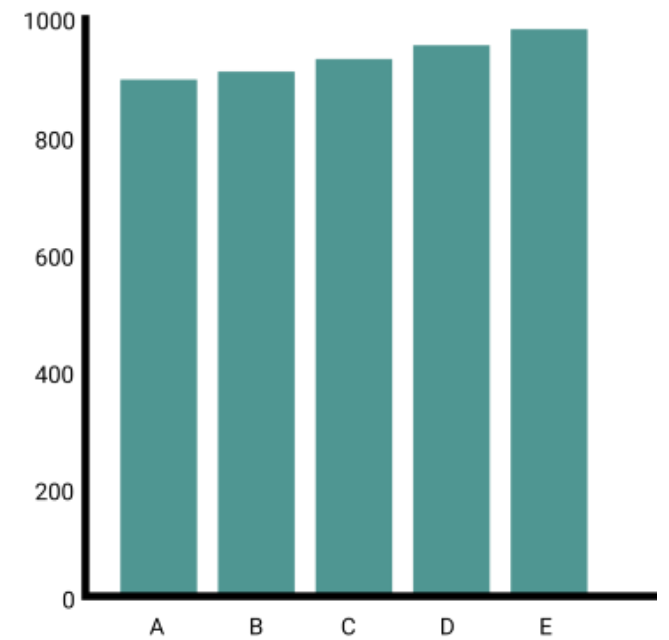


Adding Zero-Break

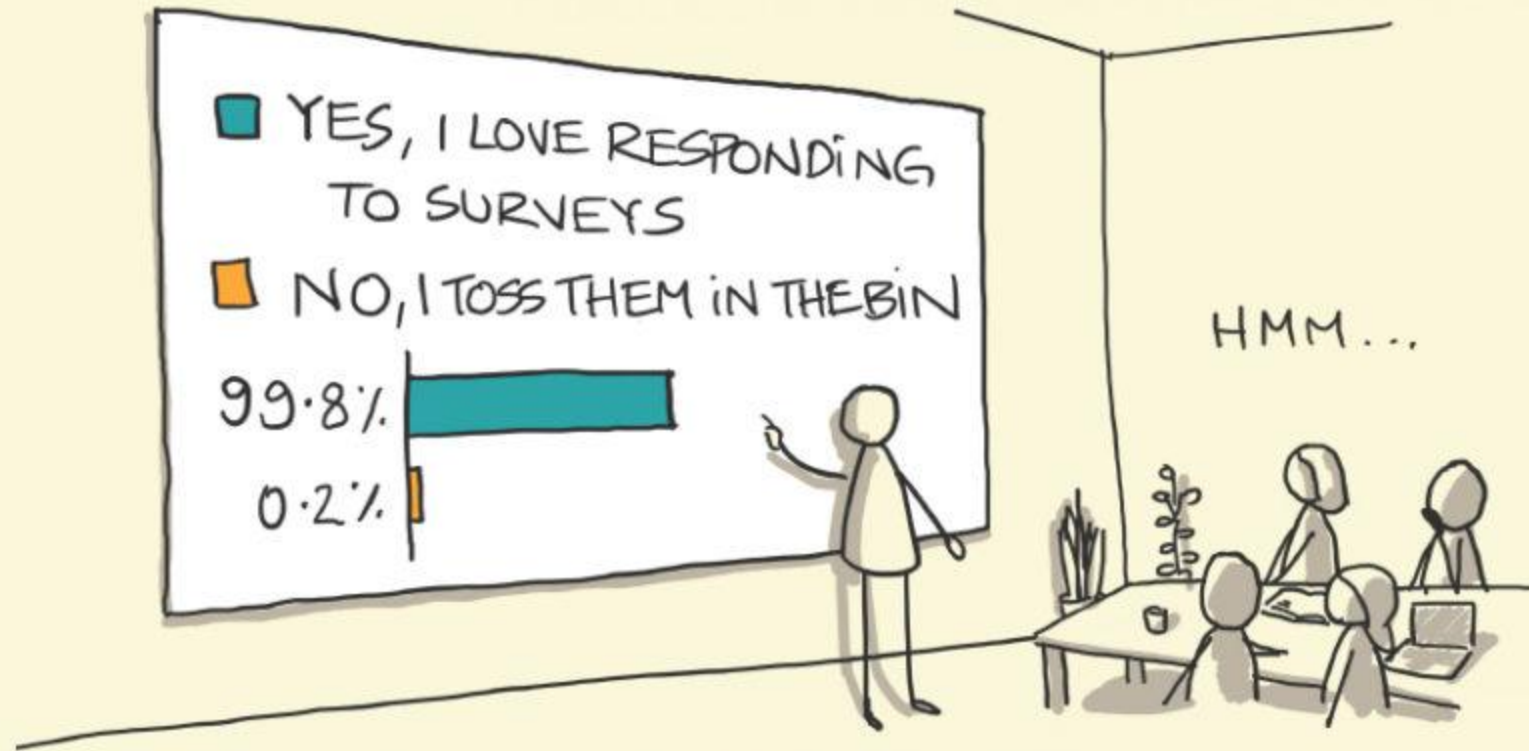


or

Starting from Zero



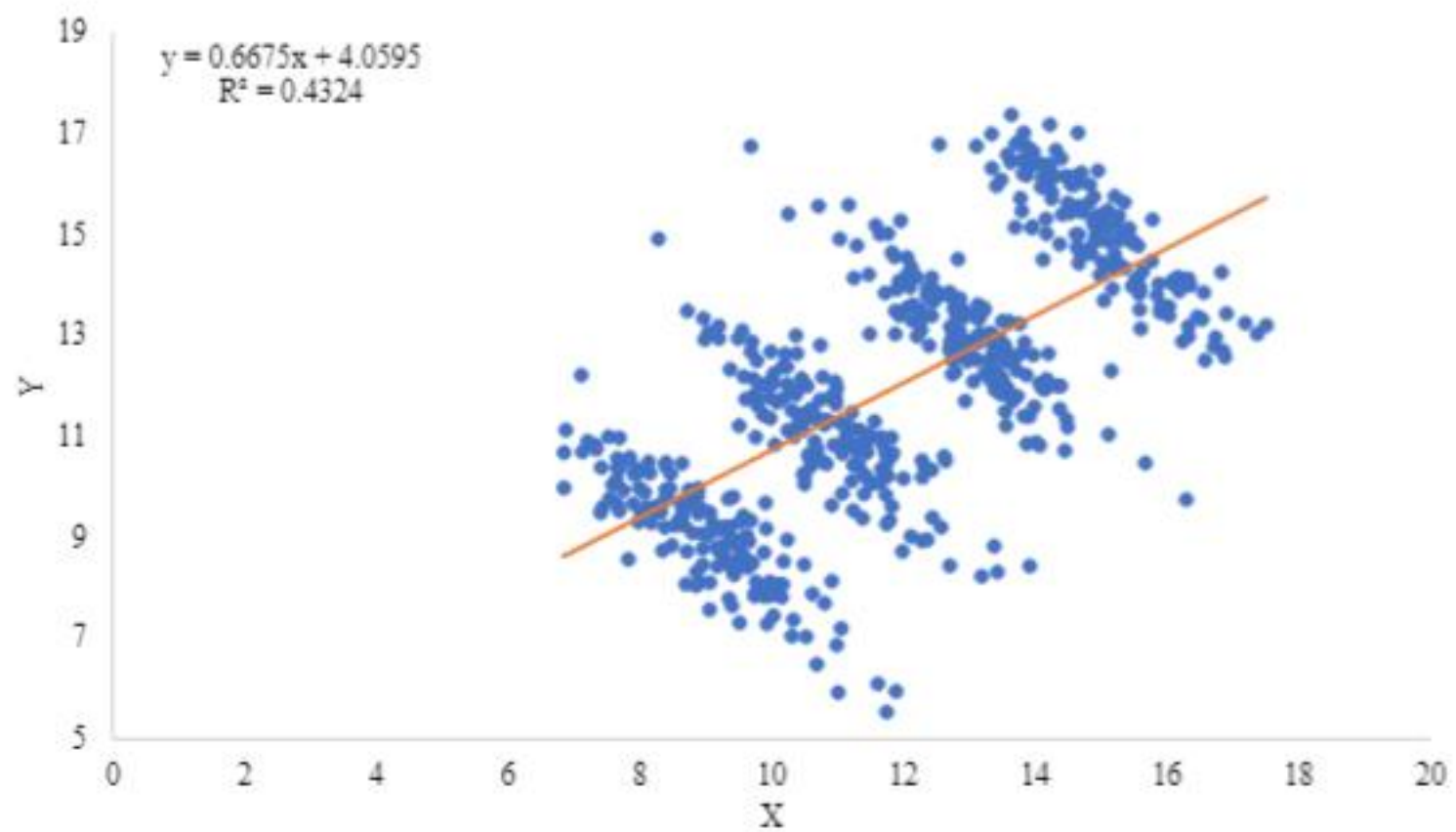
SAMPLING BIAS

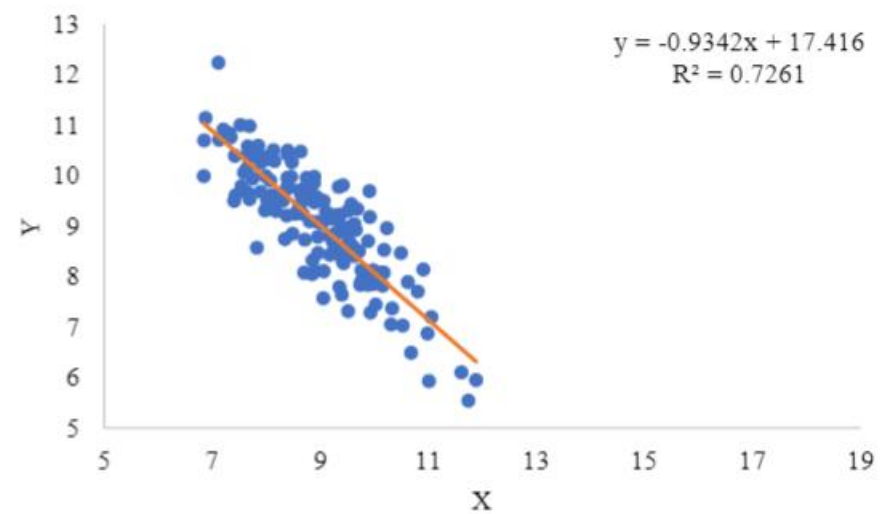
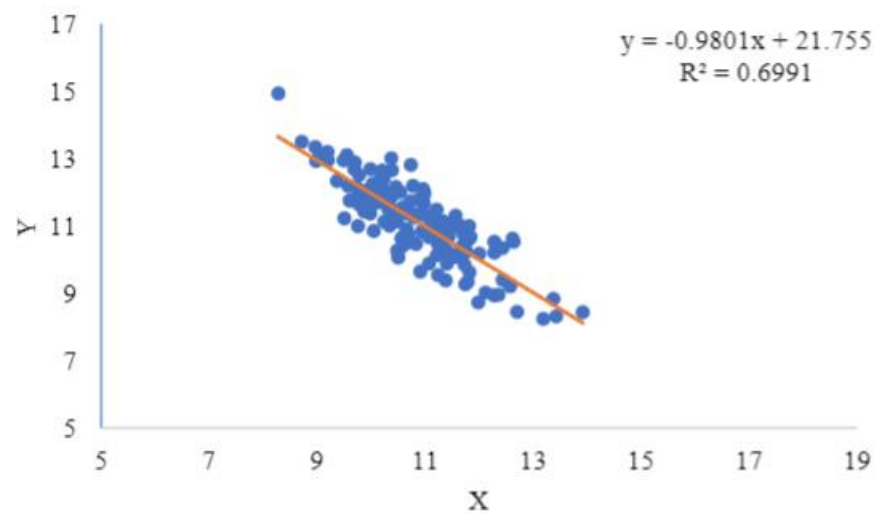
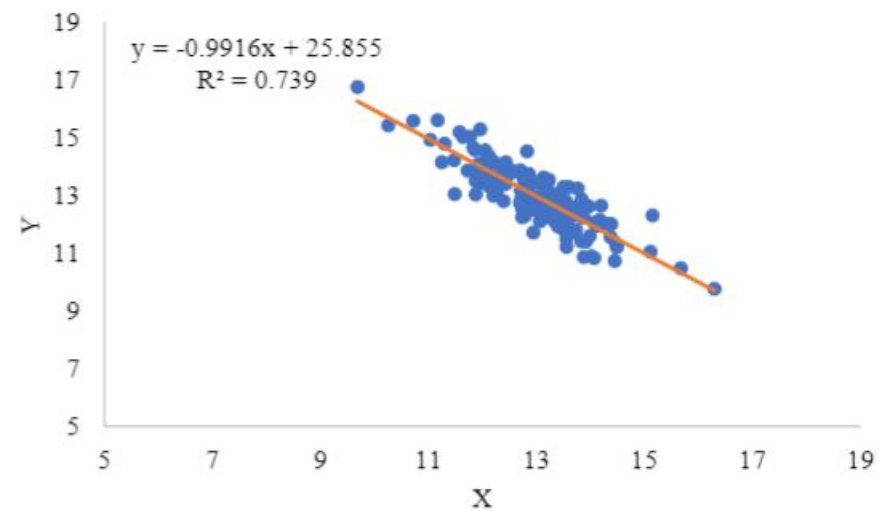
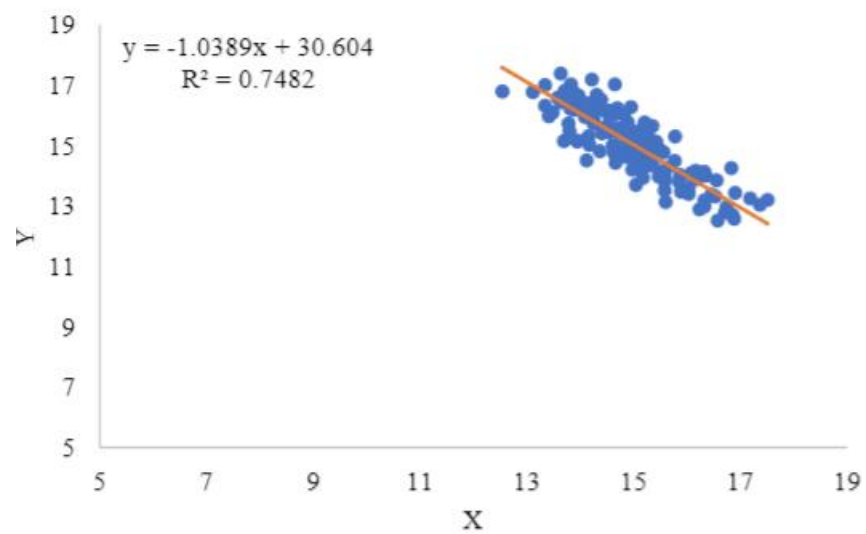


" WE RECEIVED 500 RESPONSES AND
FOUND THAT PEOPLE LOVE RESPONDING
TO SURVEYS "

STATISTICAL PARADOXES

- Simpson's paradox [20]
- Berkson's paradox [21]
- Lord's paradox [22]
- Base rate fallacy [23]
- Will Rogers paradox [24]
- Accuracy paradox [25]
- Braess's paradox [26]





THE FAMOUS EXAMPLE (GENDER BIAS IN GRADUATE ADMISSIONS IN BERKELEY 1973)

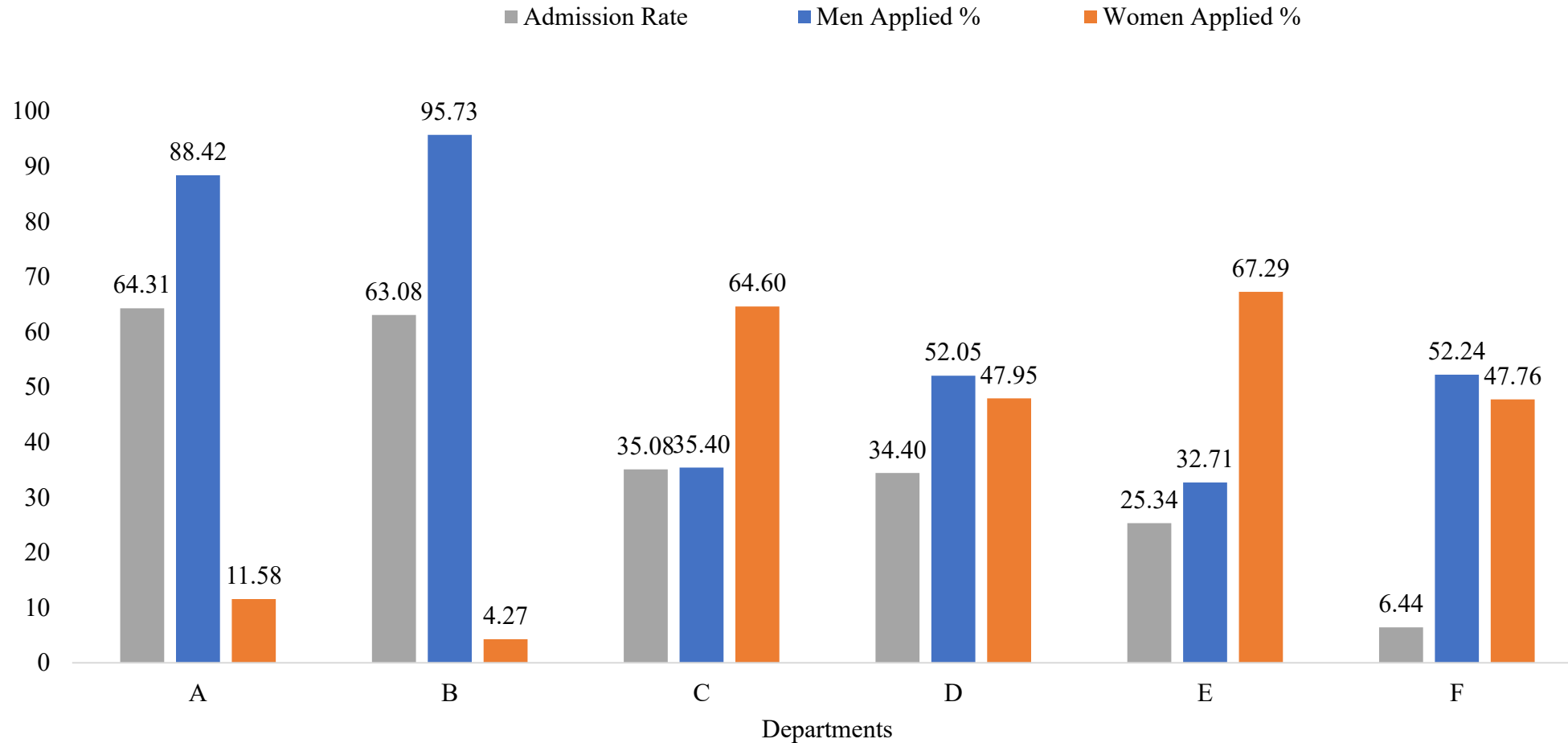
	Men		Women	
	Applicants	Admitted	Applicants	Admitted
Total	8442	44%	4321	35%

Department	Men		Women	
	Applicants	Admitted	Applicants	Admitted
A	825	62%	108	82%
B	560	63%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%

- Women have significantly better admission rates than men in most departments.
- Overall, women have a significantly smaller admission rate than men.

Bickel, Peter J., Eugene A. Hammel, and J. William O'Connell. "Sex bias in graduate admissions: Data from Berkeley." *Science* 187.4175 (1975): 398-404.

THE FAMOUS EXAMPLE (GENDER BIAS IN GRADUATE ADMISSIONS IN BERKELEY 1973)



- Why? Women would rather apply to departments with significantly lower admission rates.

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